

Lab 1 - Prelab

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Reading Assignment

1. Chapters 1 and 2 in Papen.
2. Chapters 1 and 2 in Collins.

Theory Problems

Spectrum of AM modulated signals

Let $s(t)$ be a bandlimited baseband signal with a frequency content $S(f)$ given by $1 - |f|/f_{\max}$, where f is the frequency and $f_{\max}$ is the maximum frequency of the baseband signal. This baseband signal is multiplied by $\cos(2\pi f_c t)$ to produce an amplitude-modulated passband signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$, where f_c is the carrier frequency and $f_c \gg f_{\max}$.

1. Sketch the amplitude of the frequency spectrum of the baseband signal $S(f)$.
2. Sketch the amplitude of the frequency spectrum of the amplitude-modulated passband signal $\tilde{S}(f)$.
3. The AM signal is demodulated by multiplying $\tilde{s}(t)$ by a coherent signal of the form $\cos(2\pi f_c t)$. The signal is filtered by an ideal lowpass filter with a cutoff frequency $f_{\max}$. Sketch the amplitude of the frequency spectrum of the demodulated signal.

4. *Frequency demodulation errors*

Here we consider the effect of a frequency error in the demodulation. An AM signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$ is demodulated using $\cos(2\pi f_c [1 + x]t)$ where x is a relative frequency error, and $s(t)$ is given in Problem 1. Sketch the amplitude of the frequency spectrum of the demodulated signal for: (a) $x = 0$, (b) $x = f_{\max}/(10f_c)$, and (c) $x = f_{\max}/f_c$. Comment on the results. Does a frequency demodulation error affect the Nyquist rate of the baseband signal? Explain.

5. *Phase demodulation errors*

An AM signal $\tilde{s}(t) = A \cos(2\pi f_c t)$ where A is a constant is demodulated by $\cos(2\pi f_c t + \phi)$ where ϕ represents a phase error.

- a) Determine an expression for the demodulated signal as a function of the phase error ϕ .
- b) The demodulated signal is now integrated over a time period $T = 1$ that is many times the period of the carrier. What is the maximum phase error ϕ that can be tolerated for the demodulated signal to ensure that the amplitude is within ten percent of the amplitude without a phase error?

Matlab Simulations

We will be mostly use Matlab (and/or) the Python numerical tools for this class. However, feel free to use any software suite with appropriate signal processing toolboxes. The directions for the simulations use standard Matlab functions.

1. *Spectral Analysis using the FFT*

Generate a sinusoid time-series vector $x(n)$ at frequency $f_c = 256\text{Hz}$, sampled at rate $f_s = 8192\text{Hz}$.

$$x(n) = x(nT_s) = A \cos(2\pi f_c n T_s)$$

- a) Using the generated time-series $x(n)$, apply the 32-point FFT to create the vector $X[k]$.

$$X[k] = \text{FFT}\{x(n)\}$$

- b) Generate the power sequence of $X[k]$ denoted $P[k] = (X[k])^2$. You can do this by typing $\text{abs}(X[k])^2$.
- c) Plot $P[k]$ vs frequency in hertz. (Note that the frequency axis from left to right corresponds to $f \in [0, 2\pi]$ in normalized frequency units)
- d) Plot the properly frequency shifted version of $P[k]$ using the matlab function $\text{fftshift}(\cdot)$. (This will use a frequency axis with limits $f \in [-\pi, \pi]$ in normalized frequency units).
- e) Using your results from part (d), correctly label the frequency axis using non-normalized frequency units in hertz. (i.e. the two delta impulses corresponding to the Fourier transform of a sinusoid should lie on frequencies -256Hz and $+256\text{Hz}$).
- i. What is the frequency spacing in hertz between each point of the FFT output?
- f) Using your results from part (e), plot the power spectrum in decibel scale using the $\text{db}(\cdot)$ function.

$$\text{db}(x) = 10 \log_{10}(x)$$

2. *Power Spectrum*

Following the steps from Simulation problem 1, generate and plot the power spectra (on a decibel scale) of the following three waveforms using at least 1024 points:

- a) *Random Sequence of -1's and +1's (BPSK - Binary Phase Shift Keying)*

First, generate a random vector of -1's and +1's:

- i. By using the function $\text{randsrc}(1, 128)$. This will create a random vector of -1's and +1's of size 128.
- ii. Convert the random sequences of -1's and +1's into rectangular pulses of 8 samples wide all with the same value by using the command $\text{kron}(x, \text{ones}(1, 8))$. You should now have a vector of size 1024.

- b) *Random Sequence of 0's and 1's (ASK - Amplitude Phase Shift Keying)*

Modify your time-series from part (a) to be a random sequence of 0's and 1's.

- c) *Additive "White" Gaussian Noise (AWGN)*

Gaussian random vector with unit variance. (Use the function $\text{randn}(1, 1024)$).

- d) Compare all three power spectra from parts (a)-(c).